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PROJECT PROPOSAL

OOP 101 – OBJECT-ORIENTED PROGRAMMING

Computer Science Department

1st Semester, AY 2026-2027

I. TEAM INFORMATION

Team Name:	An Family		
Team Number:	4	Section:	CS 2101

TEAM MEMBERS			
No.	SR-Code	Name	GitHub Username
1	25-00030	Carandang, Sean Christ Trez' Davidoff A.	carandangsean27-bot
2	25-04981	Palma, Tristan P.	TheVectorGod
3	25-04407	Palor, Christian Gene C.	Yula-0o0

II. PROJECT INFORMATION

Project Title

ECPC: Efficient Custom Personal Computer Builder

Project Background

Building a custom personal computer is a technical task that can be difficult for people who are unfamiliar with computer hardware. Components such as the processor, motherboard, graphics card, memory, and power supply must work together, and choosing among the many available models is not straightforward. Buying incompatible parts wastes money and time, since the parts may have to be returned or replaced.

Budget is another major concern. Each component must be balanced against the others and matched to its cost, because the most expensive parts do not always make the most suitable build for a given purpose. For example, a powerful graphics card paired with a weak processor leaves part of its performance unused. Users may also prefer



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certain brands, such as Intel or AMD processors or specific graphics card manufacturers, which narrows their options further.

To address these problems, the Efficient Custom Personal Computer Builder (ECPC) is proposed. ECPC is a Java-based console application that helps users design a PC build based on their budget, intended use, and brand preferences. It is efficient in two ways: it recommends the best performance for the user's budget, and it saves users from checking part compatibility by hand. The system works from a catalog of PC components stored in a data file, with each part's specifications, performance score, and price. It checks compatibility between the selected parts, estimates power requirements, calculates the total cost, rates the balance between the processor and graphics card, and compares components by their performance-to-price value.

Project Objectives

General Objective

To develop a Java-based application that guides users in designing a compatible PC build within their budget by validating component compatibility and power requirements, rating the balance between the processor and graphics card, and recommending parts based on the user's intended use.

Specific Objectives

The project aims to:

1. Build a parts catalog of CPUs, motherboards, GPUs, RAM, storage, power supplies, and cases, loaded from a data file with each part's specifications, performance score, and price.
2. Implement compatibility checks for CPU–motherboard socket, RAM type and capacity, case form factor and GPU length, and power supply wattage with a safety margin.
3. Implement an auto build feature that recommends a complete, compatible build within the user's minimum and maximum budget based on workload and optional brand preference, and reports when no feasible build exists.
4. Implement a bottleneck detector that rates the CPU–GPU balance as balanced, moderate, or severe for the chosen workload.
5. Allow users to create, edit, save, and load custom builds.



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6. Apply object-oriented programming principles in the design so that new component types can be added with minimal changes to existing code.

Target Users

Target User	How They Will Use the System
Students	Students who need a computer for schoolwork such as programming, research, or editing, or for light gaming, on a limited budget. They enter a budget range and intended use, and ECPC recommends compatible parts that fit.
First-Time PC Buyers	Buyers who want to build a PC but do not know which parts to choose or which parts work together. They provide their budget, intended use, and brand preference, and ECPC recommends compatible parts and warns about incompatibilities or performance mismatches.
PC Hobbyists	Users who already have a part list in mind. They enter their chosen parts through Custom Build to check compatibility, power requirements, and balance, then save the build.

Proposed Features

Feature	Description
Compatibility Check	Verifies the CPU–motherboard socket, RAM type and capacity limit, case form factor and GPU length, and that the power supply wattage covers the estimated power draw with a safety margin.
Bottleneck Detector	Compares CPU and GPU performance scores for the chosen workload and flags the limiting part when the mismatch is moderate or severe, whether the CPU or the GPU is the weaker part.
Auto Build and Budget Optimizer	Generates a complete build within the user’s minimum and maximum budget for the chosen workload (schoolwork, gaming, or video editing) and optional



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	brand preference, selecting parts with the best performance-to-price value.
Custom Build	Lets users add, replace, or remove specific parts for a build they have in mind, with the running total cost and compatibility status shown as they edit.
Save and Load Builds	Stores a build in a file and loads it later for viewing or editing.

III. JAVA AND OOP CONCEPTS

OOP Concept	How It Will Be Used
Classes and Objects	Define blueprints for the hardware and the system logic: Component, CPU, Motherboard, GPU, RAM, Storage, PowerSupply, PCCase, PCBuild, PartCatalog, and others. Objects represent specific catalog parts with their own specifications, the user’s current build, and generated reports.
Methods	Encapsulate core operations such as calculating total cost and estimated wattage (PCBuild), running compatibility checks (CompatibilityChecker), computing CPU–GPU balance (BottleneckAnalyzer), and generating a recommended build (BuildRecommender).
Encapsulation	Attributes such as price, socket type, wattage, dimensions, and performance score are private and accessed through public getters. Catalog parts have no setters, so they cannot change after loading. Changes to a build go through methods such as addComponent(), which validate the input before updating the build.
Inheritance	The abstract class Component holds the shared attributes (ID, name, brand, price). CPU, Motherboard, GPU, RAM, Storage, PowerSupply, and PCCase extend it and



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	add their own specifications, which avoids repeated code.
Polymorphism	All parts are handled as Component objects (for example, stored in one list and displayed in one loop), while each subclass overrides methods such as getSpecs() and checkCompatibility() with its own behavior. A CPU checks its socket against the motherboard, while a GPU checks its length against the case.
Abstraction	Component is an abstract class that declares methods every part must implement. Calculations such as wattage totals and bottleneck formulas sit behind simple method calls, so the menu code only calls them and displays the results.
Exception Handling	Custom exceptions (such as IncompatibleComponentException and BudgetExceededException) and try-catch blocks handle invalid input like letters entered as a budget, incompatible parts added to a build, budgets with no feasible build, and missing or corrupted catalog and save files, without crashing the program.
Arrays/Data Handling	ArrayList stores the parts catalog, the parts in the current build, and the list of saved builds.

IV. PROPOSED CLASSES

Class	Purpose	Main Attributes	Main Methods
Component (abstract)	Base class of all hardware parts.	id, name, brand, price	getSpecs() (abstract), checkCompatibility(build) (abstract), getters
CPU, Motherboard, GPU, RAM, Storage, PowerSupply, PCCase	Subclasses of Component with part-specific	CPU: socket, tdp, performanceScore	Overridden getSpecs() and checkCompatibility(build); getters



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	specifications and rules.	Motherboard: socket, ramType, maxRamGb, formFactor GPU: tdp, lengthMm, performanceScore RAM: ramType, capacityGb Storage: type, capacityGb PowerSupply: wattage PCCase: supportedFormFactors, maxGpuLengthMm	
PartCatalog	Loads and stores all available parts.	parts (ArrayList of Component)	loadFromFile(), getPartsByType(), findById(), filterByBrand()
PCBuild	The user's current build. Also what BuildRecommender returns.	buildName, selected CPU, motherboard, RAM, GPU, storage, power supply, and case	addComponent(), removeComponent(), getTotalPrice(), getEstimatedWattage(), displaySummary(), clearBuild()
CompatibilityChecker	Checks a whole build: runs each part's own check plus the build-level power check.	(none stored; works on a PCBuild)	checkAllParts(build), checkPowerSufficiency(b uild), getCompatibilityReport(b uild)
BottleneckAnalyzer	Rates the CPU-GPU balance for a workload.	cpuScore, gpuScore, workload, mismatchPercentage, bottleneckLevel	calculateMismatch(), getBottleneckLevel(), getBottleneckReport()
BuildRecommender	Generates an auto build within the budget range (budget optimization).	minBudget, maxBudget, workload, preferredBrand	generateBuild(), selectBestValuePart(), getRecommendedBuild()
Workload (enum)	Defines the intended use and how it affects part selection.	SCHOOLWORK, GAMING, VIDEO_EDITING;	getBudgetShare(), getCpuWeight(), getGpuWeight()

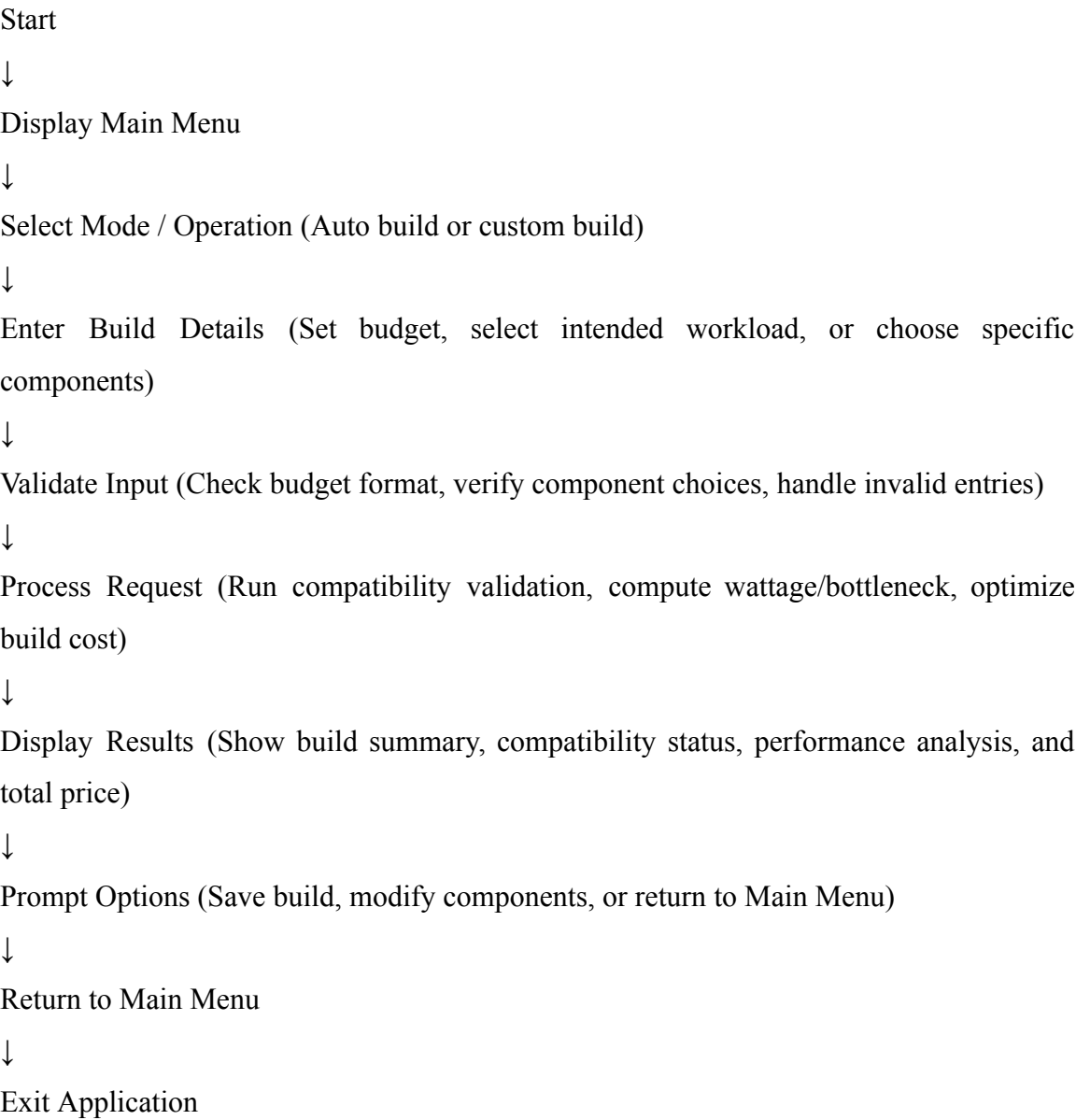


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		budget share per part; CPU and GPU weights	
BuildFileManager	Saves and loads builds from a file.	filePath	saveBuild(), loadBuild(), listSavedBuilds()
Custom exceptions	IncompatibleComponentException, BudgetExceededException, and InvalidInputException signal incompatible parts, over-budget requests, and invalid user input to the menu code.		
ECPCApp (main class)	Runs the console menu and handles user input.	catalog, current build, scanner	run(), showMainMenu(), handleAutoBuild(), handleCustomBuild()

The classes may be changed as the project is developed.

V. PROGRAM FLOW





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VI. TEAM RESPONSIBILITIES

Member	Assigned Tasks
Carandang, Sean Christ Trez’ Davidoff A.	Lead Programmer: Coordinates the implementation of Java classes, methods, and OOP features.
Palma, Tristan P.	Project Leader: Coordinates tasks, monitors deadlines, communicates with the instructor, and facilitates team meetings.
Palor, Christian Gene C.	System/Design Lead: Develops system requirements, class designs, UML diagrams, and overall application structure.

VII. DEVELOPMENT SCHEDULE

Week	Activity	Expected Output
Week 1	Planning and requirements	Project proposal
Week 2	System and class design	Class/UML diagram
Week 3	Initial coding	Basic classes
Week 4–5	Feature development	Working features
Week 6	OOP implementation	Completed OOP features
Week 7	Testing and debugging	Tested application
Week 8	Documentation and presentation	Final project

The schedule may be adjusted based on the course calendar.



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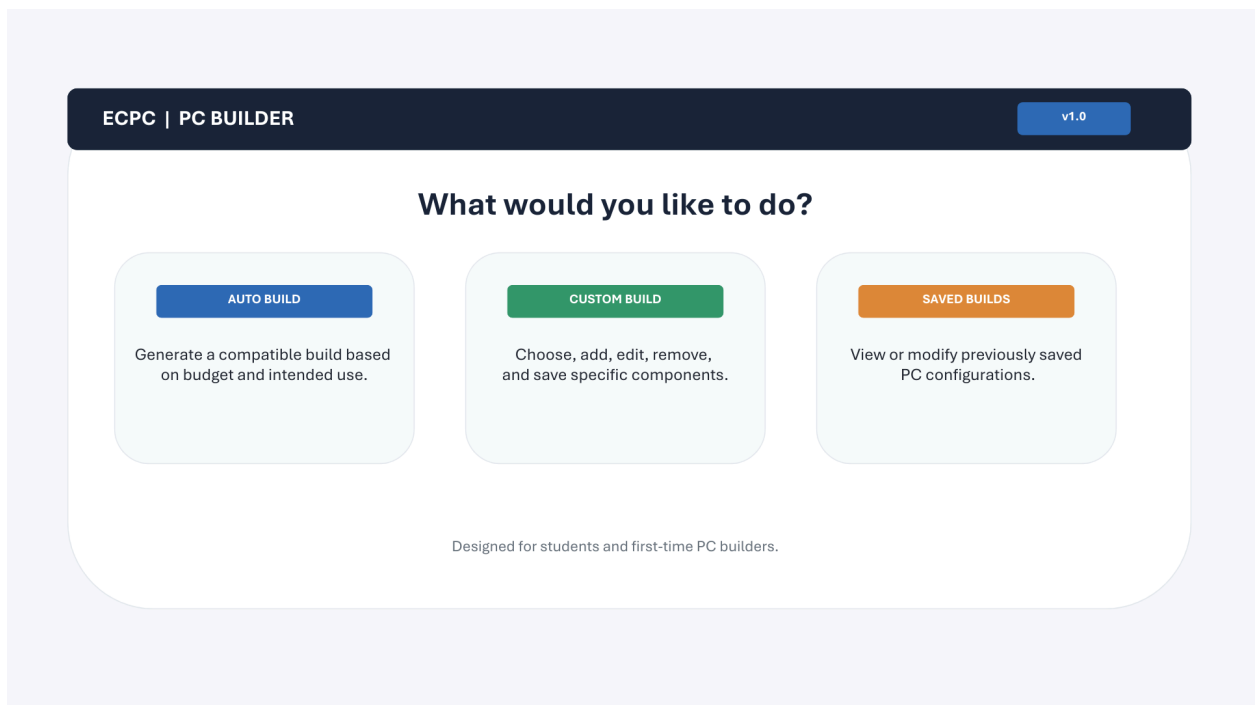
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VIII. EXPECTED OUTPUT

The team expects to produce a functional Java application that helps users plan a custom PC build based on their budget, intended use, and brand preference. The application will recommend a compatible build within the user’s minimum and maximum budget, let users create, edit, save, and load their own builds, and show each build’s compatibility status, estimated power requirement, CPU–GPU bottleneck rating, and total price. These results help users avoid incompatible parts and unnecessary expenses. The application will demonstrate the appropriate use of Java and object-oriented programming concepts.

Example UI implementation:

MAIN MENU UI





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AUTO BUILD UI

Auto Build Setup

Collect the minimum information needed for the recommendation algorithm.

Build Requirements

Budget	P50,000
Intended Use	Gaming
Priority	Balanced / Value
Brand Preference	Any

What ECPC will evaluate

- ✓ Available component data
- ✓ Compatibility rules
- ✓ Performance-to-price value
- ✓ Budget constraints
- ✓ CPU-GPU balance
- ✓ PSU power requirement

GENERATE BUILD

GENERATED BUILD SUMMARY UI

Generated PC Build

The main result: selected components, prices, and key performance data.

Recommended Build

WITHIN BUDGET

COMPONENT	MODEL	PRICE	PERFORMANCE	STATUS
CPU	Ryzen 5 7600	P10,500	CPU Mark: 27,xxx	✓
Motherboard	B650M Gaming Plus	P8,500	—	✓
GPU	Radeon RX 7600	P16,500	GPU Mark: 19,xxx	✓
RAM	32GB DDR5 6000	P5,500	—	✓
Storage	1TB NVMe SSD	P3,500	—	✓
PSU	650W 80+ Bronze	P3,500	650W	✓
Case	Airflow mATX Case	P2,500	GPU clearance	✓

TOTAL COST: P50,000

Why these parts? Best compatible value within budget.



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COMPATIBILITY CHECK AND POWER ANALYSIS UI

Compatibility & Power Analysis

The application explains whether the generated configuration can work together.

Compatibility Check

CPU ↔ Motherboard Socket	PASS
RAM ↔ Motherboard	PASS
Motherboard ↔ Case	PASS
GPU ↔ Case Clearance	PASS
Storage ↔ Motherboard	PASS
PSU ↔ System Power	PASS

Power Analysis

Estimated system load

≈ **380 W**

Selected PSU

650 W

SUFFICIENT

Power validation uses the component data and rules stored in the local dataset.

PERFORMANCE EVALUATION AND BUDGET ASSESSMENT UI

Performance & Budget Analysis

Turn raw component data into understandable recommendations.

CPU-GPU Balance

GOOD

Relative performance comparison

Value / Price

GOOD

Benchmark score + reference price

Budget Status

Build is within target budget

ECPC ASSESSMENT

The recommended configuration satisfies the budget and compatibility rules, while providing a balanced CPU-GPU combination for the selected workload.

Note: performance and balance indicators are based on the project's documented dataset and transparent rules.



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IX. PROJECT LIMITATIONS

- No live pricing, store integration, purchasing, checkout, or payment processing. Part specifications, performance scores, and prices come from a predefined local dataset, so only parts in the dataset can be recommended and prices may be out of date.
- Performance scores and bottleneck ratings are estimates based on fixed reference scores. The system does not run benchmarks, diagnostics, or overclocking tools on physical hardware and cannot predict exact performance in a specific game or application.
- Compatibility rules cover standard desktop parts only: CPU, motherboard, GPU, RAM, storage, power supply, and case. CPU coolers, case fans, custom liquid cooling, peripherals, laptops, pre-built systems, and specialized server hardware are not supported.
- ECPC is an application with no user accounts, or online features. Saved builds are stored in a local file.

X. GITHUB REPOSITORY

Repository Name: ECPC-Efficient-Custom-Personal-Computer-Builder

Link toRepository:

<https://github.com/AN-FAMILY-OOP101/ECPC-Efficient-Custom-Personal-Computer-Builder.git>



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XI. APPROVAL

Prepared by:

Name	Signature
Carandang, Sean Christ Trez' Davidoff A.	
Palma, Tristan P.	
Palor, Christian Gene C.	

Instructor's Comments:

Project Status:

- ☐ Approved
☐ Approved with Revisions
☐ Revise and Resubmit

Name of Instructor:

Ms. FATIMA MARIE P. AGDON, MSCS

Date Signed: _____